

Name: Key

10. a) Compute an orthonormal basis for the subspace spanned by this set of vectors in $\mathbb{R}^4$:

$$\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 6 \\ 3 \\ 0 \\ -1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 3 \\ 0 \\ -1 \end{pmatrix} = 6 + 3 + 0 - 1 = 8 \quad \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = 4$$

$$\begin{pmatrix} 6 \\ 3 \\ 0 \\ -1 \end{pmatrix} - \frac{8}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \\ 0 \\ -1 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ -2 \\ -3 \end{pmatrix}$$

So $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right\}$ form an orthogonal basis for the subspace.

Now: rescale to normalize.

$$\left\| \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\| = \sqrt{4} = 2$$

$$\left\| \begin{pmatrix} 4 \\ 1 \\ -2 \\ -3 \end{pmatrix} \right\| = \sqrt{16 + 1 + 4 + 9} = \sqrt{30}$$

$$\rightarrow \left\{ \begin{pmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{pmatrix}, \begin{pmatrix} 4/\sqrt{30} \\ 1/\sqrt{30} \\ -2/\sqrt{30} \\ -3/\sqrt{30} \end{pmatrix} \right\} \text{ is an orthonormal basis for the subspace.}$$

b) What is a basis for the subspace of $\mathbb{R}^3$ that is orthogonal to v ?

$$v = \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}$$

Set $(2 \ 1 \ -4)$ to be the row space of a matrix. Then the nullspace of the matrix is the orthogonal complement to $\text{span}\left\{\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}\right\} \subseteq \mathbb{R}^3$.

$$(2 \ 1 \ -4) \xrightarrow{1/2 R_1} \begin{pmatrix} 1 & 1/2 & -2 \end{pmatrix} \quad \text{so} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -1/2 \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$$

So $\left\{ \begin{pmatrix} -1/2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $(\text{span}\{v\})^\perp$.

other answers are possible,
like $\left\{ \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$

Name: Key

10. a) Compute an orthonormal basis for the subspace spanned by this set of vectors in $\mathbb{R}^3$:

$$\left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$$

$$\begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = -2 + 0 + 0 = -2, \quad \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = 1 + 0 + 9 = 10$$

$$\begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} - \frac{-2}{10} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \frac{1}{5} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix}$$

So $\left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix} \right\}$ is an orthogonal basis for the subspace.

rescaling $\rightarrow \left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 9/5 \\ 3 \\ 3/5 \end{pmatrix} \right\} \rightarrow \left\{ \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\} \rightarrow \boxed{\left\{ \begin{pmatrix} -1/\sqrt{10} \\ 0 \\ 3/\sqrt{10} \end{pmatrix}, \begin{pmatrix} 3/\sqrt{35} \\ 5/\sqrt{35} \\ 1/\sqrt{35} \end{pmatrix} \right\}}$

orthonormal basis

b) Give a basis for the subspace of $\mathbb{R}^4$ that is orthogonal to v .

$$v = \begin{pmatrix} 2 \\ 1 \\ 3 \\ -4 \end{pmatrix}$$

norm = $\sqrt{10}$ norm = $\sqrt{35}$

$$\left\{ \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \frac{1}{\sqrt{35}} \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\} \text{ or } \left\{ \frac{1}{\sqrt{10}} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \frac{1}{\sqrt{35}} \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right\}$$

Set $(2 \ 1 \ 3 \ -4)$ to be the row space of

a matrix. Then the nullspace of the matrix is the orthogonal complement to $\text{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ 3 \\ -4 \end{pmatrix} \right\} \subseteq \mathbb{R}^4$

$$(2 \ 1 \ 3 \ -4) \xrightarrow{1/2 R_1} \begin{matrix} x & y & z & w \\ 1 & 1/2 & 3/2 & -2 \end{matrix}$$

$$\text{So } \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = y \begin{pmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + z \begin{pmatrix} -3/2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + w \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad \text{So } \left\{ \begin{pmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3/2 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

is a basis for $(\text{span}\{v\})^\perp$.